

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 96-166

WASTE DISCHARGE REQUIREMENTS
FOR

LAKE COUNTY SPECIAL DISTRICTS
SOUTHEAST REGIONAL WASTEWATER SYSTEM
LAKE COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Lake County Special Districts, Southeast Regional Wastewater System (hereafter Discharger) submitted a Report of Waste Discharge, dated 6 March 1996 for proposed changes to its wastewater treatment and disposal facility.
2. Waste Discharge Requirements Order No. 88-009, adopted by the Board on 29 January 1988, prescribes requirements for a discharge from an oxidation ditch activated sludge facility to an effluent storage reservoir followed by pasture irrigation disposal.
3. Lack of effluent storage capacity and disposal capacity has caused overflows from the effluent storage reservoir during periods of above average rainfall. These surface water discharges are in violation of Order No. 88-009 and Lake County Ordinances. Cease and Desist Order No. 91-202 was adopted 21 February 1991 with a time schedule for the District to obtain adequate disposal facilities. The proposed changes by the Discharger are in response to Cease and Desist Order No. 91-202.
4. The Discharger proposes to construct a modified aerated pond system to treat the wastewater prior to disinfection and disposal. The existing oxidation ditch system will be abandoned.
5. The sewage treatment plant accepts wastewater from homes and commercial businesses in the Clearlake and Lower Lake areas. The Discharger is considering accepting leachate from the East Lake Landfill in Clearlake. The leachate is characterized as typical municipal landfill leachate which has high BOD and low dissolved metals.
6. There exists a known geothermal resource area (KGRA) in the vicinity of Cobb Mountain approximately 14 miles southwest of the current facility. The KGRA has been actively tapped for steam production, in order to generate electricity, since the late 1970s. Steam production has decreased in recent years. Injection of water into the steamfield has proven to increase production from specific areas.

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7. The Discharger, in partnership with geothermal steam suppliers and users, proposes to construct a 26 mile pipeline to deliver treated and disinfected wastewater and supplemental lake water to the KGRA for injection. Injection of the water will prolong the life of the steamfield and thus create electricity. An agreement for the disposal of wastewater has been signed by all parties.
8. The Discharger currently treats and disposes up to 4.2 million gallons per day during wet weather to existing on-site disposal facilities.
9. Wastewater effluent from the treatment system is referred to as effluent and the blend of effluent and lake water contained in the storage reservoir and in the transmission pipeline is referred to as injection water.
10. The Discharger proposes to use the storage reservoir at the Middletown Sewage Treatment Plant for temporary storage of injection water for emergencies and scheduled maintenance. The Middletown Sewage Treatment Plant is also owned and operated by Lake County Special Districts.
11. The treatment plant is in Sections 9, 10, 15, and 16, T13N, R7W, MDB&M, with surface water drainage to Burns Valley Creek, which is tributary to Clear Lake, as shown in Attachment A, which is attached hereto and part of the Order by reference.
12. The ultimate disposal area for the injection water is in the KGRA, commonly referred to as the Southeast Geysers Area in various sections within Township 11 North and Range 8 West, MDB&M. The Delivery Point, the point at which the water users become responsible for handling and disposal of the water, is in Section 32, T11N, R8W, MDB&M. The Delivery Point is defined as the first pump station in Bear Canyon (BC1). Surface water drainage from the disposal area generally flows to Putah Creek which is tributary to Lake Berryessa as shown on Attachment B, which is attached hereto and part of the Order by reference.
13. Some of the injection wells to be used by the geothermal suppliers are in the Russian River Hydrologic Basin which is regulated by the North Coast Regional Water Quality Control Board.
14. The Board adopted a Water Quality Control Plan, Third Edition, for the Sacramento River and San Joaquin River Basins (hereafter Basin Plan), which contains water quality objectives for all waters of the Basin. These requirements implement the Basin Plan.
15. The beneficial uses of Clear Lake and Cache Creek are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.

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16. The beneficial uses of Lake Berryessa and Putah Creek are municipal, industrial, and agricultural supply; recreation; aesthetic enjoyment; navigation; ground water recharge; fresh water replenishment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
17. The beneficial uses of the Russian River are determined by the North Coast Regional Water Quality Control Board, and are municipal and domestic water supply, agricultural water supply, industrial service and process supply, ground water recharge, navigation, hydropower generation, contact/noncontact water recreation, warm and cold fresh water habitat, wildlife habitat, and fish migration and spawning.
18. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.
19. The Basin Plan encourages reclamation. Injection of blended effluent, and lake water, for steam/electric production is an encouraged reclamation project.
20. Lake County has certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.), and the State CEQA Guidelines. As stated in the EIR, the project may have the following significant and/or potentially significant impacts on water quality:
 - a. Construction of the project includes stream crossings and trenching which may lead to siltation of creeks and water ways, or raise the turbidity in areas near the construction site.
 - b. Failure of the Geysers Effluent Pipeline could result in a spill of injection water and related wash-out at the failure point.
 - c. The Geysers Effluent Pipeline could experience slow leaks that could contaminate local ground water.
21. The Board has reviewed the EIR. These waste discharge requirements will mitigate or avoid significant or potentially significant impacts on water quality listed in Finding 20. Specifically, the impacts will be mitigated as follows:
 - a. Construction of the project will be done under applicable storm water construction permits, stream bed alteration permits, lake bed alteration permits, and water quality certifications.
 - b. Impacts caused by failure of the Geysers Effluent pipeline will be minimized through the use of isolation valves at approximately two mile intervals, and automatic monitoring equipment to shut down pump stations in the event of pipeline failure. Impacts from failures will also

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be mitigated through effluent limits for the Southeast Regional Wastewater Treatment Plant effluent in the pipeline.

- c. Ground water contamination caused by slow leaks from the pipeline will be mitigated by imposing effluent limitations contained in these waste discharge requirements, and by prohibiting releases of injection water from the pipeline. Additionally, the Discharger proposes to encase the pipeline where its alignment passes within 50 feet of domestic wells.
22. This discharge is exempt from the requirements of Title 23, CCR, Section 2510, et seq. (hereafter Chapter 15). The exemption, pursuant to Section 2511(b), is based on the following:
- a. The Board is issuing waste discharge requirements, and
 - b. The discharge complies with the Basin Plan, and
 - c. The effluent does not need to be managed according to 22 CCR, Division 4, Chapter 30, as a hazardous waste.
23. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
24. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Lake County Special Districts, Southeast Regional Wastewater System, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes, effluent, or injection water to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated effluent is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Chapter 15, is prohibited.
4. Discharges or releases of injection water from any segment of the pipeline, other than at the Delivery Point or storage at the Middletown Sewage Treatment Plant, is prohibited.

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B. Treatment Plant Discharge Specifications:

1. The monthly average wet weather effluent flow shall not exceed the plant capacity of 6.1 million gallons/day.
2. The maximum daily effluent flow shall not exceed the plant hydraulic capacity of 8.5 million gallons.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the treatment plant area (property owned by the Discharger).
4. As a means of maintaining compliance with Treatment Plant Discharge Specification No. 3, the Discharger shall initiate an odor investigation within one week upon receiving complaints of odor production. The Discharger shall initiate an odor mitigation study within one month of verifying that odor complaints are attributable to the plant operation.
5. The treatment facilities shall be designed, constructed, operated, and maintained to prevent inundation or washout due to floods with a 100-year return frequency.
6. The effluent from the treatment facility to the storage reservoir shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Monthly Median</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	40		80
Total Coliform	MPN/100 ml		23	500

¹ Five-day, 20° Celsius biochemical oxygen demand.

7. Ponds shall not have a pH less than 6.0 or greater than 9.0.
8. Ponds shall be managed to minimize breeding of mosquitos. A mosquito abatement program shall be submitted to this office within 120 days following adoption of this Order.
9. Public contact with effluent and injection water shall be precluded through such means as fences, signs, and other acceptable alternatives.

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10. Treatment ponds and the storage reservoir shall have sufficient capacity, in conjunction with disposal capacity, to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet for the treatment ponds (measured vertically to the lowest point of overflow).

C. Sludge Disposal:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15, Division 3, Title 23, of the California Code of Regulations and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice from a previously approved practice shall be reported to the Executive Officer and U.S. Environmental Protection Agency (EPA) Regional Administrator at least 90 days in advance of the change.
3. Use and disposal of sewage sludge shall comply with existing Federal and State laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the Regional Water Quality Control Boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

4. The Discharger is encouraged to comply with the State Guidance Manual issued by the Department of Health Services titled *Manual of Good Practice for Landspreading of Sewage Sludge*.
5. By **2 January 1997**, the Discharger shall submit a sludge disposal plan describing the annual volume of sludge generated by the plant and specifying the disposal practices, including sampling, testing, and application procedures.

D. Ground Water Limitations:

The treatment and discharge of effluent to the storage reservoir shall not cause underlying ground water to:

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1. Be degraded.
2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in 22 CCR, Division 4, Chapter 15.
3. Exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
4. Exceed concentrations of radionuclides specified in 22 CCR, Division 4, Chapter 15.
5. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.
6. Contain concentrations of chemical constituents in amounts that adversely affect agricultural use.

E. Provisions:

1. The Discharger shall comply with the Monitoring and Reporting Program No. 96-166, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements", dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. In the event of any change in control or ownership of land at the Southeast Treatment Plant or waste discharge facilities described herein (pipeline facilities), the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.
4. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
5. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may

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result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.

6. The Discharger shall submit an Emergency Response Plan by **2 January 1997** to assure compliance with Discharge Provisions A.1. and A.4. of this Order.
7. A copy of this Order shall be kept at the discharge facility for reference by operating personnel. Key operating personnel shall be familiar with its contents.
8. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.
9. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 21 June 1996.



WILLIAM H. CROOKS, Executive Officer

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CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 96-166

FOR
LAKE COUNTY SPECIAL DISTRICTS
SOUTHEAST REGIONAL WASTEWATER SYSTEM
LAKE COUNTY

Specific sample locations shall be established under direction of the Board's staff and a description of the stations shall be attached to this Order.

Prior to construction, plans and specifications for ground water monitoring wells shall be submitted to Board staff for review and approval.

EFFLUENT MONITORING

Effluent samples shall be collected just prior to discharge to the Effluent Storage Reservoir. Effluent samples should be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Sampling Frequency</u>
BOD _{5, 20°C}	mg/l	Composite	Weekly
Suspended Solids	mg/l	Composite	Weekly
Specific Conductance	μmhos/cm	Grab	Weekly
pH	pH units	Grab	Weekly
Total Coliform	MPN/100 ml	Grab	Weekly
Flow	MGD	Cumulative	Continuously

SLUDGE MONITORING

The Discharger shall monitor the sludge production and quality in accordance with the sludge production and disposal plan as approved by the Executive Officer.

WATER SUPPLY MONITORING

Representative samples of the municipal water supplies shall be obtained and tested as follows, or in lieu of water samples, test results of tests conducted by individual water suppliers covering the following compounds shall be reported:

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<u>Constituents</u>	<u>Units</u>	<u>Sampling Frequency</u>
Standard Minerals	mg/l	Annually
Specific Conductance	μ mhos/cm	Annually
Total Dissolved Solids	mg/l	Annually

GROUND WATER MONITORING

Prior to construction of wells, plans and specifications for ground water monitoring wells shall be submitted to Board staff for review and approval.

REPORTING

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner to illustrate clearly the compliance with waste discharge requirements.

Monthly monitoring reports shall be submitted to the Regional Board by the **20th day of the following month.**

The results of any monitoring done more frequently than required at the locations specified in the Monitoring and Reporting Program shall be reported to the Board.

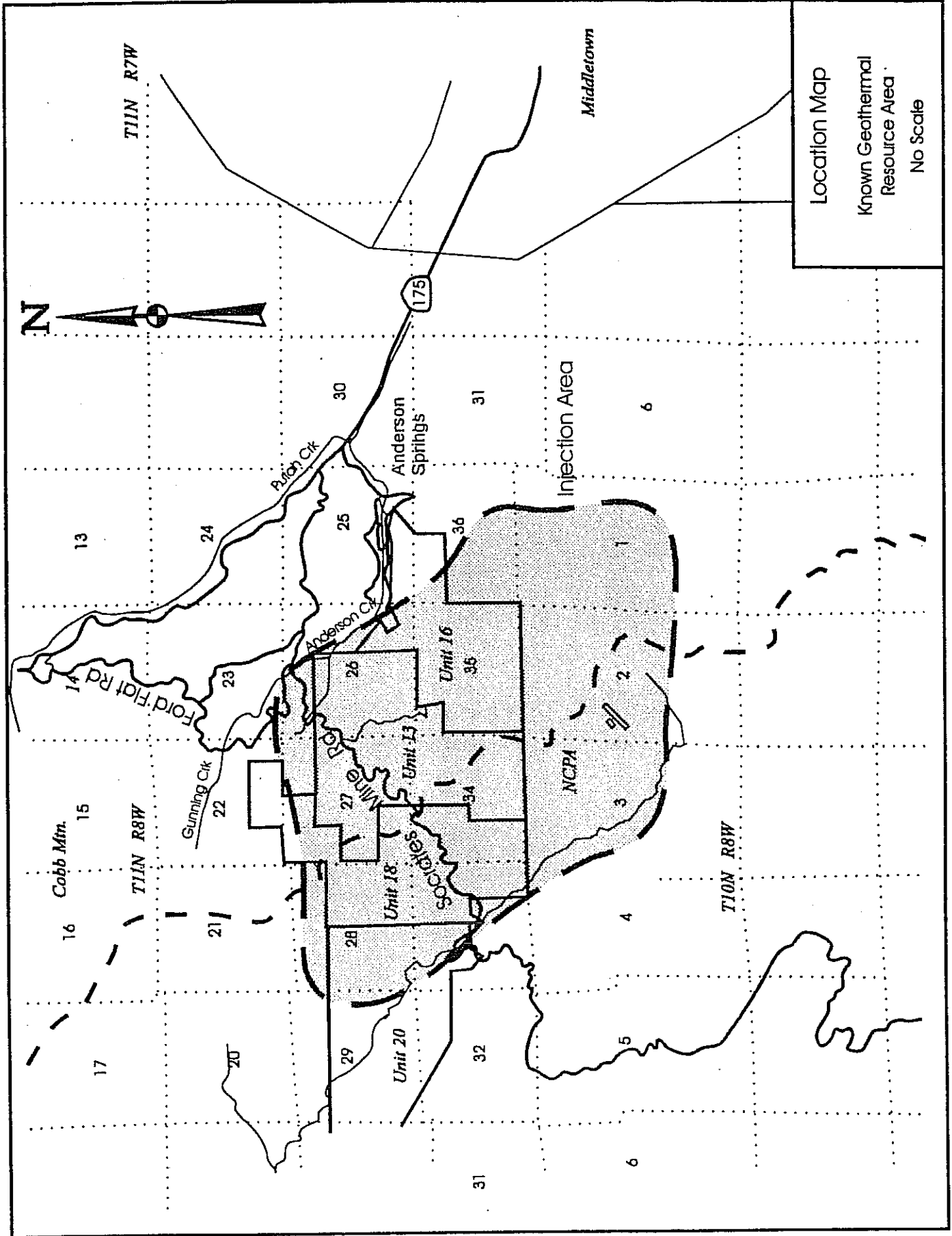
Upon written request of the Board, the Discharger shall submit a report to the Board by **30 January** of each year. The report shall contain both tabular and graphical summaries of the monitoring data obtained during the previous year. In addition, the Discharger shall discuss the compliance record and the corrective actions taken or planned which may be needed to bring the discharge into full compliance with the waste discharge requirements.

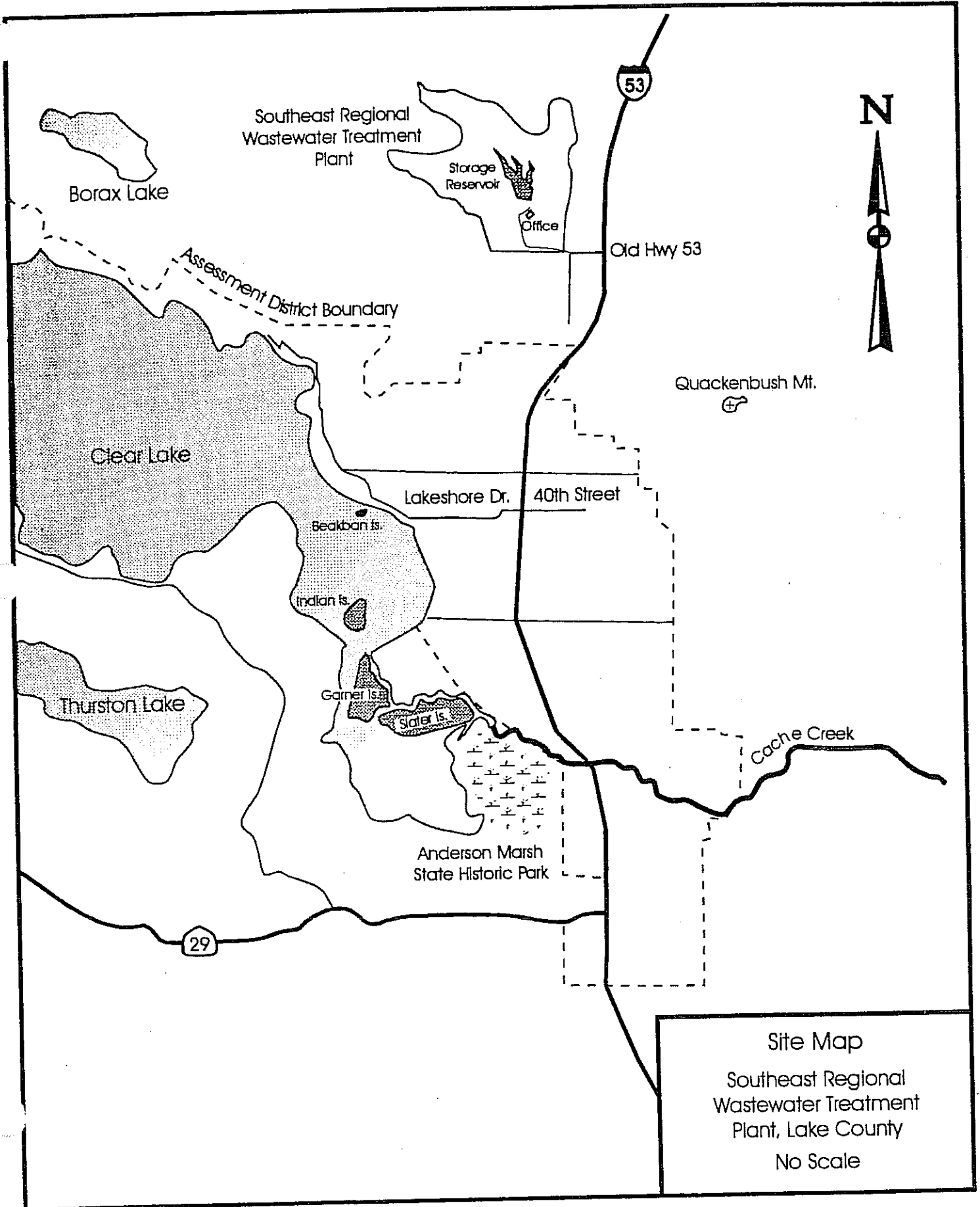
The Discharger shall implement the above monitoring program as of the date of this Order.

Ordered by : William H. Crooks
WILLIAM H. CROOKS, Executive Officer

21 June 1996
(Date)

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INFORMATION SHEET

LAKE COUNTY SPECIAL DISTRICTS SOUTHEAST REGIONAL WASTEWATER SYSTEM LAKE COUNTY

The District owns and operates a municipal sewage treatment plant, serving the Clearlake and Lower Lake areas. Currently, sewage is treated in an activated sludge process. Effluent is disinfected, stored, and used for irrigation of pasture land. Lack of effluent storage capacity and disposal capacity has caused overflows from the effluent storage reservoir during periods of above average rainfall. These surface water discharges are in violation of waste discharge requirements and Lake County Ordinances. A Cease and Desist Order was adopted in 1991 with a time schedule for the District to obtain adequate disposal facilities. The proposed changes by the Discharger are in response to this Cease and Desist Order.

The District, in partnership with geothermal steam suppliers and users, proposes to construct a 26 mile pipeline to deliver treated and disinfected wastewater and supplemental lake water to the Southeast Geysers Steamfield, near Middletown, for injection. Injection of the water will prolong the life of the steam field and thus create electricity.

This Order places effluent limits on the Southeast Regional Wastewater Treatment System effluent portion of the injection water. BOD limits are proposed to minimize odor potential in the storage reservoir and to ensure good disinfection potential. Disinfection of the effluent is required to minimize impacts in the event of failure of the pipeline and also to protect workers in the geothermal area, prior to injection. The Department of Health Services does not require the lake water to be treated or disinfected, even though the lake water may contain coliform.

The District relinquishes responsibility of the injection water at the Delivery Point, defined as the Number 1 Bear Canyon Pump Station. An association of geothermal steam suppliers and steam users will become responsible for the transport and injection of the injection water following the Delivery Point. The District has proposed a series of mitigation measures to minimize environmental impacts from failures of the pipeline, such as automatic pump cut-off devices, and siphon breakers at localized high spots.

Discharge of wastewater to the Geysers is considered an encouraged reclamation project. Wastewater will be converted to steam which will in turn be converted to electricity.

Surface water drainage from these facilities are to Cache Creek and Putah Creek.